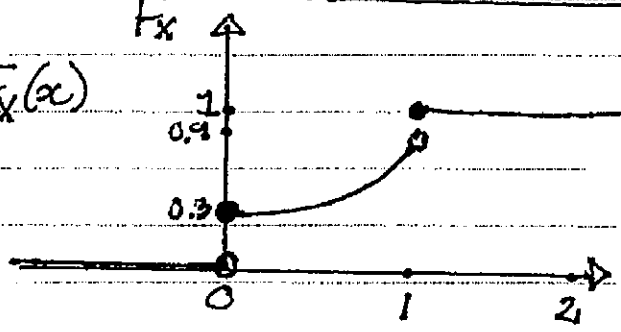
F78PA 2019-20 Semester 1 Exam Solutions

December 2019-20

① (a) Graph of $F_X(x)$



$F_X(x)$ must satisfy: ① $F_X(x) \in [0,1]$ $\forall x \in \mathbb{R}$

② $F_X(x)$ is non-decreasing.

③ $\lim_{x\to\infty} F_X(x) = 1$, $\lim_{x\to-\infty} F_X(x) = 0$

④ $F_X(x)$ is right continuous.

X is neither discrete nor continuous. It is not a step function so not discrete, but has discontinuities at 0 and 1 so not continuous.

(b) $\text{Im } X = [0,1]$,

(c) (i) $\Pr(X=1) = F_X(1) - \lim_{x\uparrow 1} F_X(x) = 1 - 0.9 = 0.1$

(ii) $\Pr(0.5<X<1) = \lim_{x\uparrow 1} F_X(x) - F_X(0.5) = 0.9 - (0.3+0.6\times 0.5^2)$

$= 0.9 - 0.45 = 0.45$

(d) $E(X) = \Pr(X=0)\times 0 + \Pr(X=1)\times 1 + \int_0^1 F_X'(x)\, x\, dx$

$= 0.1 + \int_0^1 1.2x^2 dx = 0.1 + \left[0.4x^3\right]_0^1$

$= 0.5$

Expected payout is £500 $\Rightarrow$ Minimum annual premium should be £500 – and more if company wishes to make a profit.

(2) (a) $E(X) = Pr(X=50)\times 50 + Pr(X=5)\times 5$

$= \frac{1}{6^3}\times 50 + \binom{3}{2}\cdot\frac{1}{6^2}\times\frac{5}{6}\times 5 = 0.579$

$E(X^2) = \frac{1}{6^3}\times 50^2 + \binom{3}{2}\cdot\frac{1}{6^2}\cdot\frac{5}{6}\times 25 = 13.31$

$\Rightarrow Var(X) = E(X^2) - [E(X)]^2 = 13.31 - 0.579^2 = 12.97$

(b) Profit for a single player given by $Y = X - 0.5$

$\Rightarrow E(Y) = E(X) - 0.5 = 0.079$, $Var(Y) = Var(X) = 12.97$

Let $T = \sum_{i=1}^{100} Y_i$ denote total profit for 100 players $\Rightarrow E(T) = 7.9$

$Var(T) = 1297$

$\Rightarrow SD(T) = 36.01$

By CLT $T \sim N(7.9, 36.01^2)$

$Pr(T \geq 50) \approx Pr\left(\frac{T-7.9}{36.01} \geq \frac{50-7.9}{36.01}\right) = Pr(Z \geq 1.17)$

$= 1 - \Phi(1.17)$

$= 1 - 0.879 = 0.121$

(c) Let C denote the charge.

Then for this version of the game, if X is payout

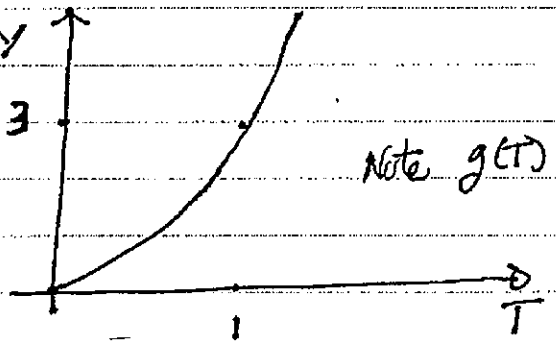
$E(X) = \frac{1}{6^3}\times 50 + \binom{3}{2}\frac{1}{6^2}\times\frac{5}{6}\times 5 + \binom{3}{1}\times\frac{1}{6}\times\frac{5^2}{6^2}\times C$

$= 0.579 + 0.347\times C$

Expected profit is zero when $C = 0.579 + 0.347\times C$

$\Rightarrow C = 0.579/0.653 = 0.89$

(3) a) $T \sim \text{Gamma}(5,1)$, $Y = 3T^{3/2} = g(T)$



$\text{Im } Y = [0, \infty)$

(b) Since $g(t) = 3t^{3/2}$, then $g^{-1}(y) = \left(\frac{y}{3}\right)^{2/3}$.
We can find pdf of Y as

$$f_Y(y) = f_T(g^{-1}(y)) \times \frac{d}{dy} g^{-1}(y), \quad y > 0$$

$$= \frac{1}{\Gamma(5)} g^{-1}(y)^4 e^{-g^{-1}(y)} \times \frac{2}{3}(y)^{-\frac{1}{3}}\left(\frac{1}{3}\right)^{2/3}, \quad y > 0.$$

$$= \frac{1}{4!}\left(\frac{y}{3}\right)^{\frac{8}{3}} e^{-\left(\frac{y}{3}\right)^{2/3}} \times \frac{2}{3}(y)^{-\frac{1}{3}}\left(\frac{1}{3}\right)^{2/3}$$

$$= \frac{1}{36}\left(\frac{1}{3}\right)^{\frac{10}{3}} y^{\frac{7}{3}} e^{-\left(\frac{y}{3}\right)^{2/3}}, \quad y > 0 \text{ and } 0 \text{ otherwise.}$$

(c) $\Pr(Y > 15) = \Pr(3T^{3/2} > 15) = \Pr(T > \left(\frac{y}{3}\right)^{2/3})$

$$= 1 - F_T\left(\left(\frac{y}{3}\right)^{2/3}\right)$$

Q4

(a) $f_X(x) = \begin{cases} 0, & x<0 \\ 2e^{-2x}, & x\geq 0 \end{cases}$ $\quad f_{Y|X}(y|x) = \begin{cases} 0, & y<0 \\ xe^{-xy}, & y\geq 0 \end{cases}$

Maginal distribution of X is $Exp(2)$, conditional $Y|X=x$ is $Exp(x)$.

Joint density: $f_{X,Y}(x,y) = f_X(x) f_{Y|X}(y|x) = 2e^{-2x} \cdot xe^{-xy}$

$= 2xe^{-(2+y)x}$, $x,y \geq 0$,

X & Y are not independent as $Y|X$ depends on the value of X.

(b) Since $Y|X \sim Exp(x)$ it follows that $P(Y\leq y|X=x) = 1-e^{-xy}$

$\Rightarrow P(Y>y|X=x) = e^{-xy}$

Using Partition Theorem

$$P(Y>y) = \int_0^\infty f_X(x) P(Y>y|X=x)\,dx$$

$$= \int_0^\infty 2e^{-2x} \cdot e^{-xy}\,dx = 2\int_0^\infty e^{-(2+y)x}\,dx$$

$$= \frac{2}{(2+y)}, \quad y>0$$

(c) From (b) we have $P(Y\leq y) = 1 - \frac{2}{(2+y)} = F_Y(y)$

$\Rightarrow f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{2}{(2+y)^2}$

It follows that $f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} = \frac{2xe^{-(2+y)x}}{2/(2+y)^2}$

$= (2+y)^2 x e^{-(2+y)x}$, $x>0$

$\Rightarrow X|Y=y \sim$ Gamma$(2, 2+y)$

(5) (a) Number of HWU players who draw after HWU is in $\{0,2,4,6\}$ $\Rightarrow$ Number who don't is in $\{1,3,5,7\}$.

(b) $X = \sum_{i=1}^{7} I_i$

(c) $E(I_i) = P_r(i \text{ does not draw after HWU player})$

$= \frac{8}{15}$ $\left[\frac{\#\text{ non HWU}}{\#\text{ possible opponents for } i}\right]$

Since $I_i^2 = I_i$, $E(I_i^2) = \frac{8}{15}$.

$E(I_iI_j) = P_r(i \text{ and } j \text{ do not draw HWU}) = P_r(i \text{ draw non HWU}) \times P_r(j \text{ draw non HWU} \mid I_i = 1)$

$= \frac{8}{15} \times \frac{7}{13} = \frac{56}{195}$

(d) $E(X) = E\left(\sum_{i=1}^{7} X_i\right) = \sum_{i=1}^{7} E(X_i) = 7 \times \frac{8}{15} = \frac{56}{15}$

$E(X^2) = E\left(\left(\sum_{i=1}^{7} X_i\right)^2\right) = E\left(\sum_{1 \le i,j \le 7} I_iI_j\right) = \sum_{1 \le i,j \le 7} E(I_iI_j)$

$= 7 \times E(I_1^2) + 7 \cdot 6\, E(I_1I_2)$

$= 7 \times \frac{8}{15} + 42 \times \frac{56}{195} = 15.79$

(5) (a) MGF of X_1 given by

$$M_{X_1}(\theta) = E(e^{\theta x}) = \int_0^\infty f_X(x) e^{\theta x} dx$$

$$= \int_0^\infty \lambda e^{-\lambda x} e^{\theta x} dx$$

$$= \int_0^\infty \lambda e^{-(\lambda-\theta)x} dx$$

$$= \left[\frac{\lambda e^{-(\lambda-\theta)x}}{(\theta-\lambda)} \right]_0^\infty = \frac{\lambda}{(\lambda-\theta)}$$

as required.

(b) Since the $X_1, \ldots, X_n$ are independent, identically distributed

$$M_Z(\theta) = \left(M_{X_1}(\theta)\right)^n = \left(\frac{\lambda}{\lambda-\theta}\right)^n$$

$\Rightarrow Z \sim \text{Gamma}(n, \lambda)$

(c) (i) CGF defined to be $K_Y(\theta) = \log M_Y(\theta)$

To obtain μ_Y, σ_Y^2: $\mu_Y = K_Y'(\theta)\big|_{\theta=0}$, $\sigma_Y^2 = K_Y''(\theta)\big|_{\theta=0}$

(ii) For r.v. Z in part (b)

$$K_Z(\theta) = \log M_Z(\theta) = n \log\left(\frac{\lambda}{\lambda-\theta}\right) = n(\log\lambda - \log(\lambda-\theta))$$

$$K_Z'(\theta) = \frac{n}{\lambda-\theta}, \quad K_Z''(\theta) = \frac{n}{(\lambda-\theta)^2}$$

$\Rightarrow \mu_Z = K_Z'(0) = \frac{n}{\lambda}$, $K_Z''(0) = \frac{n}{\lambda^2} = \sigma_Z^2$.

(d) Chebyshev's inequality: $\Pr(|X| \geq \delta) \leq \frac{E(X^2)}{\delta^2}$

(e) $E((Z-5)^2) = \text{Var}(Z) = \frac{10}{4}$, $\Pr(Z \geq 10) = \Pr(|Z-5| \geq 5) \leq \frac{10}{4 \times 25} = 0.1$

(7)

(a) $Cov(X,Y) = E((X-\mu_x)(Y-\mu_y))$

$$\rho(X,Y) = \frac{Cov(X,Y)}{\sigma_x \sigma_y}$$

If $\rho(X,Y) = 1$, then $Y = aX+c$ where $a>0$
If $\rho(X,Y) = -1$ $Y = aX+c$ $a<0$.

More generally $\rho(X,Y)$ captures the strength of the linear relationship between X and Y. Range: $-1 \le \rho(X,Y) \le 1$.
$Cov(X,Y) = 0 \not\Rightarrow X, Y$ independent.

(b) $Var(X+Y) = E(((X-\mu_x)+(Y-\mu_y))^2)$

$$= E((X-\mu_x)^2 + 2(X-\mu_x)(Y-\mu_y) + (Y-\mu_y)^2)$$
$$= E((X-\mu_x)^2) + 2E((X-\mu_x)(Y-\mu_y)) + E((X-\mu_x)^2)$$
$$= \sigma_x^2 + 2\rho(X,Y)\sigma_x\sigma_y + \sigma_y^2$$

This variance is monotone increasing in ρ. Positive correlation increases the variance of $X+Y$ while negative correlation reduces it.